

JUN WEI NG

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EXPERIENCE

- AI Engineer Intern

May 2026 – Aug 2026

Keppel

Singapore

 - Built an enterprise knowledge platform that identifies authoritative engineering information across document revisions using ontology, lineage, and supersedence modelling, reducing fact-verification time by ~80%.
 - Developed provider-agnostic document extraction and retrieval pipelines using Azure AI Foundry and PaddleOCR, with visual attribution tracing generated answers to exact source pages, figures, and tables.
 - Engineered Azure infrastructure spanning Blob Storage, Cosmos DB, App Service, GitHub Enterprise CI/CD, network controls, and cost monitoring, taking the division's first AI/SWE workflow from prototype to an enterprise-ready PoC.
- Data Science Intern

Jan 2026 – May 2026

GovTech

Singapore

 - Built an edge-cloud wildlife monitoring pipeline that processed 1,000+ hours of footage into searchable event clips using real-time detection, segmentation, species recognition, and multimodal captioning.
 - Developed an open-vocabulary video analytics system with MOE for detecting configurable school incidents using multimodal models and synthetic evaluation videos; the prototype received a Judges' Choice award at GovTech BUILD.
 - Containerized and benchmarked GPU inference services across cloud and edge environments, supporting real-time video streams and scalable batch processing for operational deployment within GovTech Platform AI.
- Data Scientist Intern

Jun 2025 – Aug 2025

TicTag

Singapore

 - Built multimodal retail analytics pipelines combining detection, pose estimation, segmentation, depth estimation, transcription, and speaker diarization to reconstruct customer interactions from video and audio for client deployments.
- Climate Models Analytics Intern

May 2024 – Aug 2024

Defence Science and Technology Agency (DSTA)

Singapore

 - Applied SRGAN/RDN super-resolution models and automated weather-data pipelines to improve rainfall-intensity analysis from satellite imagery.

EDUCATION

- National University of Singapore (NUS)

Aug 2025 – Aug 2026

Master of Science in Business Analytics (GPA: 4.88/5.00)

Singapore

 - DSO-AISG Incentive Award recipient; researched fine-tuning Qwen2.5-VL-7B for fine-grained biomechanical assessment from monocular exercise videos.
- National University of Singapore (NUS)

Aug 2022 – May 2025

B.Eng. Civil Engineering, Honours (Highest Distinction) (GPA: 4.67/5.00)

Singapore

 - Minors in Computer Science and Data Analytics; Specialisation in Digitalization of Urban Infrastructure; dissertation on reinforcement-learning-based autonomous mobility-on-demand rebalancing and traffic optimization.

SOFTWARE PROJECTS

- DIVA | Python, FastAPI, React, Docker, Azure, Cosmos DB

Aug 2026 – Present

 - Built an open-source document intelligence platform with visual attribution that transforms unstructured document collections into structured knowledge while tracing extracted claims to exact source regions and evidence.
 - Designed schema-driven extraction, document lineage and per-field temporal resolution, provenance-preserving retrieval, provider-agnostic OCR/LLM adapters, evaluation tooling, and containerized local and cloud deployment.
- RetailMind | AWS, Bedrock, Step Functions, Fargate, Next.js

Jan 2026 – Present

 - Built an accessible AutoML platform that guides non-technical users from business problem formulation to model training, prediction, and interpretation without requiring ML code or infrastructure management.
 - Designed asynchronous training and inference workflows using AWS Step Functions and ephemeral Fargate workers, with Bedrock translating natural-language objectives into ML tasks and business insights.

TECHNICAL SKILLS & ACHIEVEMENTS

Languages: Python, Java, SQL, JavaScript, R
ML/Data: PyTorch, Transformers, scikit-learn, pandas, NumPy, LoRA, DeepSpeed, Reinforcement Learning
Engineering: FastAPI, React, Docker, PostgreSQL, Qdrant, Neo4j, AWS, GCP, Azure
Awards: DSO-AISG Incentive Award; NUS Outstanding Undergraduate Researcher Prize; IAHR-Singapore Hydraulics Medal; Dean's List; CDE Innovation & Research Award
Leadership: President, NUS Civil Engineering Club; Resident Assistant, Residential College 4